

BSCS 3rd Semester

Probability and Statistics

Lecture #01

Introduction

Basic definitions

Random Experiment:-

Random experiment

is an action or process that can be repeated under the same action but outcome cannot be predicted with certainty.

Exp:-

Tossing a coin.. H, T.

Sample space:-

The total outcome

from any experiment is called sample space.

Exp:-

$$S = \{H, T\}$$

Event:-

A subset of sample space is called

event

Exp:-

$$S = \{HH, HT, TH, TT\}$$

$A = \{HH\}$ only head appear.

$B = \{TT\}$ only tail appear.

$C = \{HT, TH\}$ both head and tail appear.

Types of event:

There are two types of event.

i. Mutually Exclusive event

ii. Not mutually exclusive event.

Mutually exclusive event:

Two events are mutually exclusive if they are occur at the same time.

Exp:-

When a coin is tossed head or tail appear but not both.

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 3, 5\}$$

$$B = \{2, 4, 6\}$$

$$A \cap B = \phi \quad \text{These are M.A}$$

NOT mutually exclusive event:

Two events

are mutually exclusive if they cannot

Day: _____

Date: _____

occur at same time.

Exp:

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 3, 5, 6\}$$

$$B = \{2, 4, 6\}$$

$$A \cap B = \{6\}$$

Lecture #02&03

Probability:

Probability is the measure of chance or likelihood that particular event occur.

$$P(E) = \frac{n(A)}{n(S)}$$

Event
 ↓
 $n(A)$
 ↑
 $n(S)$
 sample space

Exp:

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$n(S) = 6$$

$$A = \{2, 4, 6\}$$

$$n(A) = 3$$

So,

$$P(E) = \frac{3}{6} = \frac{1}{2} = 0.5$$

mcq

∴ Probability lie between (0,1)

$$0 \leq P(A) \leq 1$$

Statistics:

It is the branch of mathematics that deal with the data to collecting, analyzing, interpreting and presenting into useful information or - making effective decision.

Exp:

Use in education to analyze student performance.

Use in business.

Data:

The collection of raw facts, numbers and observations.

Exp:

Marks of students.....

Data Analysis:

Data analyzing means studying the collected data.

Exp:

If you take the marks of student and find average you are doing data analysis.

Types of statistics:

i- Descriptive statistics

ii. Inferential statistics

Descriptive statistics:-

It is used to summarize and describe the main features of data.

Exp:

→ Mean, median, mode

Inferential statistics:-

It is used to make conclusion about a large group based on data from sample space.

Exp: If a man starts a business of chocolate he choose some people ^{from} large population for taste their chocolate and make conclusion.

Types of data:

i. Discrete data

ii. Continuous data

Discrete data:-

Countable data is called discrete data. It include only whole numbers $\{0, 1, 2, 3, \dots\}$.

Exp:

→ Number of students in a class.

Day: _____

Date: _____

Continuous data:-

Measurable data is called continuous data. It may be in decimal form.

Exp:-

- Height of a student [4.57]
- Weight of a student [60kg]

Population:-

Population means the entire group of individuals or items that you want to study.

Exp:-

- All students of a school.
- All people living in Pakistan.

Q: Probability of two dice:-

$S = \{$
 $(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$
 $(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)$
 $(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)$
 $(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)$
 $(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)$
 $(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)$
 $\}$

$$A = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$$

$$n(A) = 6$$

$$n(S) = 36$$

$$P(E) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6} = 0.16$$

Probability of two coins:

$$S = \{HH, HT, TH, TT\}$$

$$A = \{HH, HT\}$$

$$n(A) = 2$$

$$n(S) = 4$$

$$P(E) = \frac{n(A)}{n(S)} = \frac{2}{4} = \frac{1}{2} = 0.5$$



Lecture #4

Conditional Probability:-

Conditional

probability is the probability of an event occurring when another event has already occurred.

It is written as:

Day: _____

Date: _____

$P(A/B)$ Probability of A given that B.

Formula:-

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$P(A \cap B)$ = Probability when both A and B occurred

Exp:-

When 3 fair coin are tossed then find the probability.

→ At least two tail appears

→ first coin show head.

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$n(S) = 8$$

$$A = \{HTT, THT, TTH, TTT\}$$

$$n(A) = 4$$

$$P(E) = \frac{n(A)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

$$B = \{HHH, HHT, HTH, HTT\}$$

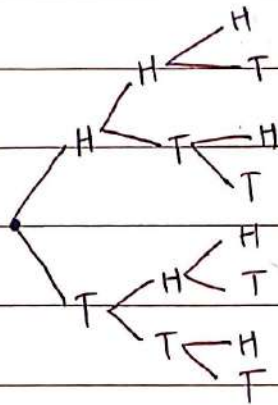
$$n(B) = 4$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

Day: _____

Date: _____

Tree method:-



Day: _____

Date: _____

Q When 4 fair coin are tossed then find the probability.

→ At least three tail appears

→ first two coin show head.

Tree method:



$S = \{ HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT, THHH, THTH, THTT, TTHH, TTHT, TTTH, TTTT \}$

$$n(S) = 16$$

$A = \{ HTTT, THTT, TTHT, TTTH, TTT \}$

$$n(A) = 5$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{5}{16} = 0.31$$

$B = \{ HHHH, HHHT, HHTH, HHTT \}$

$$n(B) = 4$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{16} = \frac{1}{4} = 0.25$$

Day: _____

Date: _____

$$A \cap B = \phi$$

$$n(A \cap B) = 0$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)}$$

$$= \frac{0}{16} = 0$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A/B) = \frac{0}{\frac{1}{4}}$$

$$P(A/B) = 0$$

Lecture #5,6

Example of Probability:

A card is drawn from an ordinary deck of 52 playing card.

→ Find probability:

Day: _____

Date: _____

1. card is Red

2. card is Diamond

3. card is 10.

Red $\rightarrow 26$

1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
K	K
Q	Q
J	J

Black $\rightarrow 26$

1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
K	K
Q	Q
J	J

 $\rightarrow 12$ Face card $\rightarrow$ Mcq's

- Red = 13 $\rightarrow$ Diamond and 13 heart • Total Red = 26
- Black = 13 $\rightarrow$ Spade and 13 club • Total Black = 26
- Face card = 12 • Total card = 52

Solution:

$$n = 52$$

$$n(S) = 52$$

① let A be the event in which card is Red

$$n(A) = 26$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{26}{52} = \frac{1}{2}$$

Day: _____

Date: _____

② Let B be the event in which card is Diamond.

$$n(B) = 13$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{13}{52} = \frac{1}{4} = 0.25$$

③ Let C be the event in which card is 10.

$$n(C) = 4$$

$$P(C) = \frac{n(C)}{n(S)} = \frac{4}{52} = \frac{1}{13} = 0.07$$

Imp Question:-

Multiplication Law/Rule:/Product Law

Statement:- If A and B are

the two events defined in sample space S then

$$\bullet P(A \cap B) = P(A) \cdot P(B|A)$$

$$\bullet P(A \cap B) = P(B) \cdot P(A|B)$$

Proof:-

→ By the def of conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \because P(B) \neq 0$$

$$P(A \cap B) = P(B) \cdot P(A|B)$$

Now,

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \because P(A) \neq 0$$

$$P(A \cap B) = P(A) \cdot P(B|A)$$

=====

Lecture #07

Random ^{change} variable / Chance variable:

A variable whose value are determined by the outcome of random experiment.

Exp:

Two coin are tossed

$$= \{ \underset{2}{HH}, \underset{1}{HT}, \underset{1}{TH}, \underset{0}{TT} \}$$

when n Numerical value assign $i = 0, 1, 2$
 $\uparrow$ $i = 0, 1, 2$

→ It is also called Stochastic variable.

Counting Sample point:

Counting Sample point divide into three rule. These rules are as follows:

Rule of Multiplication

Rule of Permutation

Rule of Combination

Rule of Multiplication:

If two experiments

m and n occurs then

$$m \times n = mn \rightarrow \text{outcomes}$$

Example:

Coin $m = H, T$

$$m = 2$$

dice $n = 1, 2, 3, 4, 5, 6$

$$n = 6$$

then by multiplication rule

$$m \times n = 2 \times 6$$

$$= 12 \text{ outcomes}$$

Rule of Permutation:

In which

order of selection does matter

$${}^n P_r = \frac{n!}{(n-r)!}$$

$n \rightarrow$ Total observation

$r \rightarrow$ Selection

Example:Member in Club $n=4$ President, Secretary, Treasure $r=3$

$${}^4P_3 = \frac{4!}{(4-3)!} = \frac{4!}{1!} = \frac{4 \cdot 3 \cdot 2 \cdot 1}{1!} = 24 \text{ Ans.}$$

Rule of Combination:

In which

order of selection does not matter

$${}_n C_r = {}^n C_r = \frac{n!}{r!(n-r)!}$$

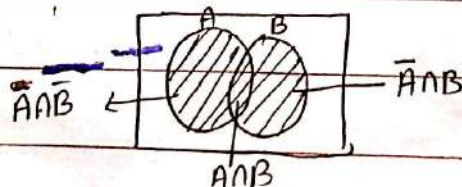
Example:List comette $n=4$ choose $r=3$

$${}_4 C_3 = \frac{4!}{3!(4-3)!}$$

$$\frac{4!}{3!(1)!} = \frac{4!}{3!1!} = \frac{24}{6} = 4$$

Additive Law:

If A and B are two not mutually exclusive events defined in S



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof:

The event $A \cup B$ is written as:

$$A \cup B = A \cup (\bar{A} \cap B)$$

$$P(A \cup B) = P(A) + P(\bar{A} \cap B) \text{ — (1)}$$

Again event B decompose

$$B = (A \cap B) \cup (B \cap \bar{A})$$

$$P(B) = P(A \cap B) + P(B \cap \bar{A}) \text{ — (2)}$$

subtract (1) and (2)

$$P(A \cup B) - P(B) = P(A) + P(\bar{A} \cap B) - P(A \cap B) - P(\bar{A} \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Lecture # 09

Discrete and Continuous Random variables:

Discrete Random variable:

→ Whole number

$$\text{Mean} = E(x) = \sum x p$$

$$\text{Variance} = E(x^2) = [E(x)]^2$$

$$\therefore E(x^2) = \sum x^2 p$$

$$\text{S.D} = \sqrt{\text{Var}(x)}$$

Example: 1:

A discrete random variable has probability given below:

$X =$	-2	-1	0	1	2	3
$P(=$	0.2	K	0.1	$2K$	0.1	$2K$

Solution:

$$0.2 + K + 0.1 + 2K + 0.1 + 2K = 1 \quad \therefore \sum p = 1$$

$$0.4 + 5K = 1$$

$$5K = 1 - 0.4$$

$$K = \frac{0.6}{5}$$

$$K = 0.12$$

$$\begin{aligned} \sum xp &= (-2)(0.2) + (-1)(0.12) + (0)(0.1) + (1)2(0.12) \\ &\quad + 2(0.1) + 3(2)(0.12) \end{aligned}$$

$$\boxed{\text{Mean} = 0.64}$$

$$\begin{aligned} \text{Variance} &= E(x^2) - [E(x)]^2 \\ &= 3.72 - (0.64)^2 \end{aligned}$$

$$\boxed{\text{Variance} = 3.3104}$$

$$\text{S.D} = \sqrt{3.3104}$$

$$\boxed{\text{S.D} = 1.819}$$

Example 2:

$x =$	0	1	2
$P =$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$$\text{Mean} = E(x) = \sum xp$$

$$= 0 + \frac{1}{4} + 1$$

$$= (0)(\frac{1}{4}) + (1)(\frac{1}{2}) + (2)(\frac{1}{4})$$

$$\boxed{E(x) = 1}$$

$$\text{Variance} = E(x^2) - [E(x)]^2$$

$$= E(x^2) = \sum x^2 p$$

$$= 1.5 - 1$$

$$= 0.5$$

$$\therefore \frac{1}{2} = 0.5$$

$$\therefore \frac{1}{2} = 0.5$$

$$\boxed{\text{Var} = \frac{1}{2}}$$

$$\text{S.D} = \sqrt{\text{Var}(x)}$$

$$\text{S.D} = \sqrt{\frac{1}{2}}$$

$$\boxed{\text{S.D} = \frac{1}{\sqrt{2}}}$$

Continuous Random Variable:

$$\text{Mean} = E(x) = \int x f(x) dx$$

$$\text{Variance} = E(x^2) - [E(x)]^2$$

$$\therefore E(x^2) = \int x^2 f(x) dx$$

Example:

$$f(x) = k(x - x^2), \quad 0 \leq x \leq 1$$

For find:

$$\text{Mean} = ?$$

$$k = ?$$

$$\text{Variance} = ?$$

Sol:

$$k \int_0^1 (x - x^2) dx$$

$$= k \left[\int_0^1 x dx - \int_0^1 x^2 dx \right] = 1$$

$$= k \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 1$$

$$k \left[\frac{(1)^2}{2} - \frac{(1)^3}{3} \right] = 1$$

$$k \left[\frac{1}{2} - \frac{1}{3} \right] = 1 \Rightarrow k \left(\frac{3-2}{6} \right) = 1$$

$$k \left(\frac{1}{6} \right) = 1 \Rightarrow \boxed{k = 6}$$

$$\begin{aligned} f(x) &= \int x f(x) dx \\ &= \int x (6(x - x^2)) dx \end{aligned}$$

$$\text{Variance} = E(x^2) - [E(x)]^2$$

$$= \left[\frac{3}{10} - \left(\frac{1}{2} \right)^2 \right]$$

$$= \frac{3}{10} - \frac{1}{4}$$

$$\boxed{\text{Var} = \frac{1}{20}}$$

Lecture #10:

Bayes theorem:

If the events $A_1, A_2, A_3, \dots, A_K$ from a partition of a sample space S , that is the events A_i are mutually exclusive and their union is S such that if it can occur only if one of the A_i occurs then for any i .

$$P(A_i/B) = \frac{P(A_i) P(B/A_i)}{\sum_{i=1}^K P(A_i) P(B/A_i)}, \text{ for } i=1, 2, \dots, K$$

Proof:

By the definition of conditional probability
 $\therefore$ Events are A_i

$$P(A_i \cap B) = P(A_i) \cdot P(B/A_i) \quad \text{--- (i)}$$

$$P(A_i \cap B) = P(B) \cdot P(A_i/B) \quad \text{--- (ii)}$$

$$P(A_i) \cdot P(B/A_i) =$$

By comparing eq (i) and (ii)

$$P(A_i) \cdot P(B/A_i) = P(B) \cdot P(A_i/B)$$

$$P(A_i/B) = \frac{P(A_i) \cdot P(B/A_i)}{P(B)} \quad \text{--- (A)}$$

We can write event B.

$$B = S \cap B$$

$$= (A_1 \cup A_2 \cup A_3 \cup \dots \cup A_k) \cap B$$

$$B = (A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_k \cap B)$$

$$P(B) = P[(A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_k \cap B)]$$

$$= P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_k \cap B)$$

∴ $A_i \cap B$ are mutually exclusive

$$P(B) = P(A_1) \cdot P(B/A_1) + \dots + P(A_k) \cdot P(B/A_k)$$

$$\sum_{i=1}^k P(A_i) \cdot P(B/A_i)$$

By putting value of $P(B)$ in eq (A),
we get

$$P(A_i/B) = \frac{P(A_i) \cdot P(B/A_i)}{\sum_{i=1}^k P(A_i) \cdot P(B/A_i)}$$

$$\sum_{i=1}^k P(A_i) \cdot P(B/A_i)$$